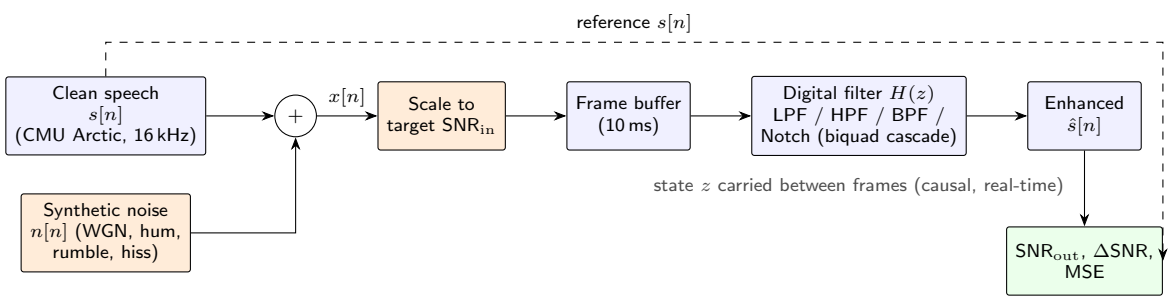
reference $s[n]$
Clean speech
$s[n]$
(CMU Arctic, 16 kHz)
Synthetic noise
$n[n]$ (WGN, hum,
rumble, hiss)
+
$x[n]$
Scale to
target $\mathrm{SNR_{in}}$
Frame buffer
(10 ms)
Digital filter $H(z)$
LPF / HPF / BPF /
Notch (biquad cascade)
Enhanced
$\hat{s}[n]$
state z carried between frames (causal, real-time)
$\mathrm{SNR_{out}}$, ΔSNR,
MSE